Eigen Value

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$$A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$$

Eigen value Formula

$\lambda^3 - [\text{Sum of Diagonal Elements}]\lambda^2 + [\text{Sum of Diagonal minors}]\lambda - |A| = 0$

Lets Solve it:-

What is Minors:

A minor is an element in a matrix is the determinant of the Smaller matrix obtained after removing the row and Column of that element.

[Sum of Diagonal Elements]

$$\begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix} = \cancel{8+9}\ 8+1-3 = 6$$

Sum of Diagonal Minors

Sum of Diagonal ~~matrix~~ Minors

minors obtained from the removing row and Column of the element

$$\begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$$

Here diagonal are 8, -3, 1 and then Sum of Diagonal minor

Cross multiplication

Minor 1 $\begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & -1 \end{bmatrix}$ $\begin{bmatrix} -3 & -2 \\ -4 & -1 \end{bmatrix}$ $-3(-1)-(-2)(-4)$

$= -3-8 = -11$

Minor 2 $\begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & -1 \end{bmatrix}$ $\begin{bmatrix} 8 & -2 \\ 3 & -1 \end{bmatrix}$ $(8\times1)-(-2\times3)$

$= 8+6 = 14$

Minor 3 $\begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & -1 \end{bmatrix}$ $\begin{bmatrix} 8 & -8 \\ 4 & -3 \end{bmatrix}$ $(8\times-3)-(-8\times4)$

$-24+32 = 8$

$= 14+8-11$

$= 22-11 = 11$

$\lambda^3 - [\text{Sum of Diagonal Elements}]\lambda^2 + [\text{Sum of Diagonal Minors}]\lambda - |A| = 0$

$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\lambda = 1, 2, 3$ [Eigen Values]

Eigenvector:
A vector whose direction remains unchanged after a linear transformation.

Eigenvalue
A scaler that represent how much the eigenvector is scaled during the transformation

Eigenvector — direction that does not Change
Eigenvalue — how much the vector is str